Infrastructure Réseau Multi-Site

Segmentation VLAN, routage OSPF, sécurité et administration à distance

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Rapport d'activité — Projet portfolio
Simulation Cisco Packet Tracer — Siège (HQ) + Agence distante (BRANCH)

1. Architecture générale

Infrastructure simulant un siège (HQ) et une agence distante (BRANCH), reliés par OSPF, avec segmentation VLAN de part et d'autre.

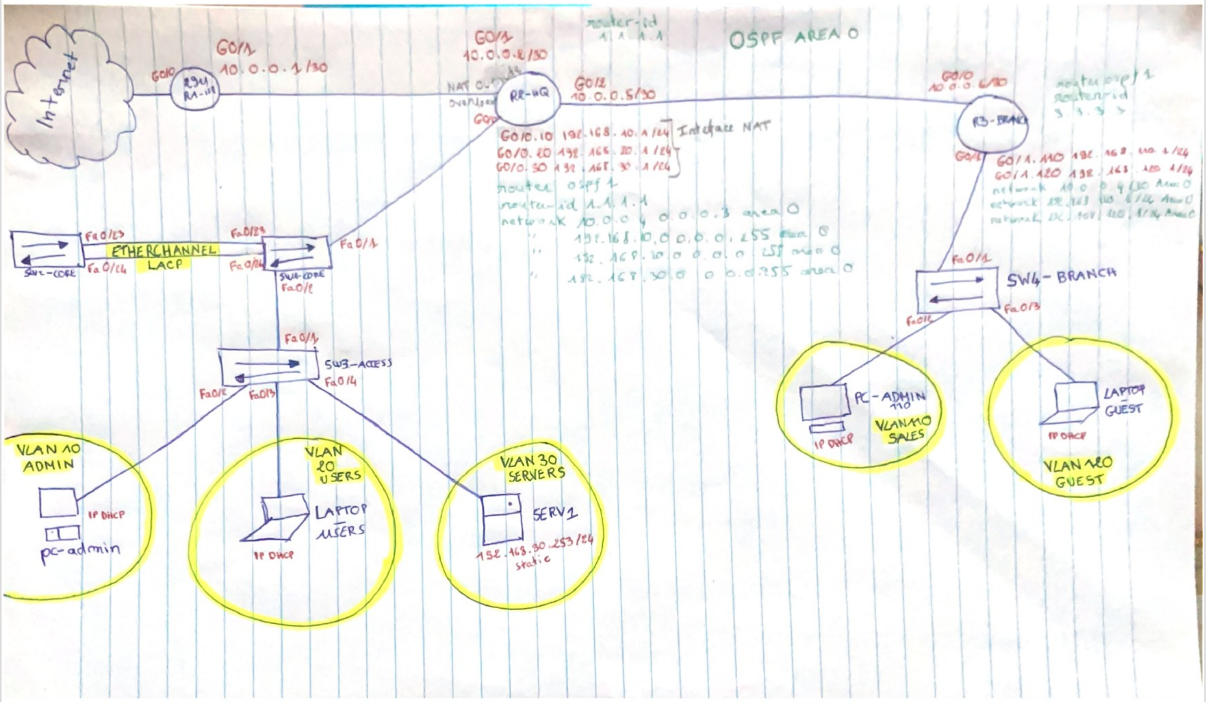
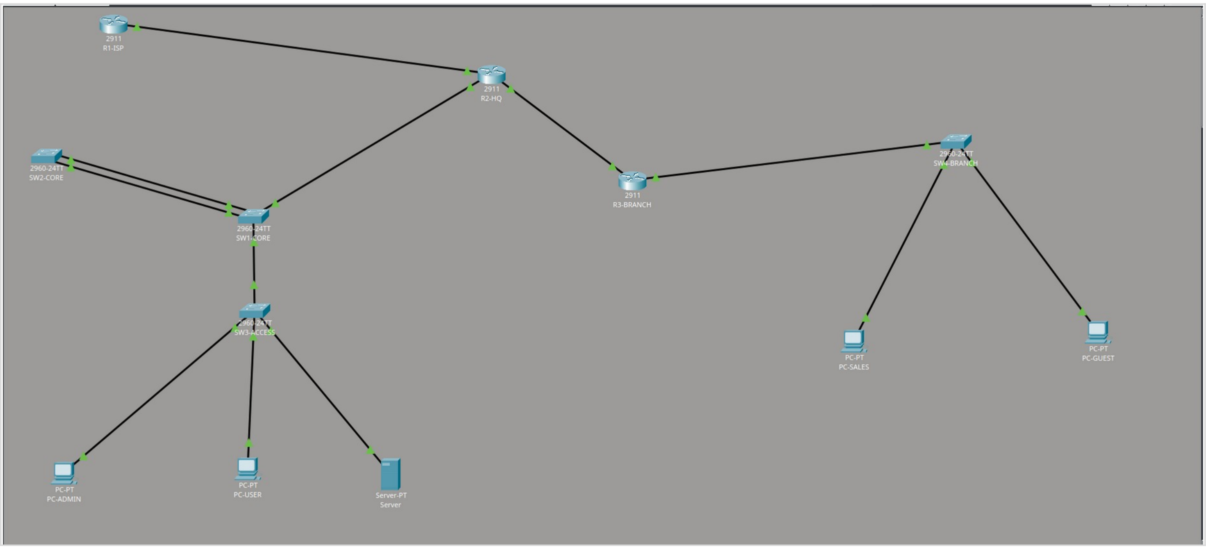


Schéma d'architecture — adressage, VLAN et interfaces.

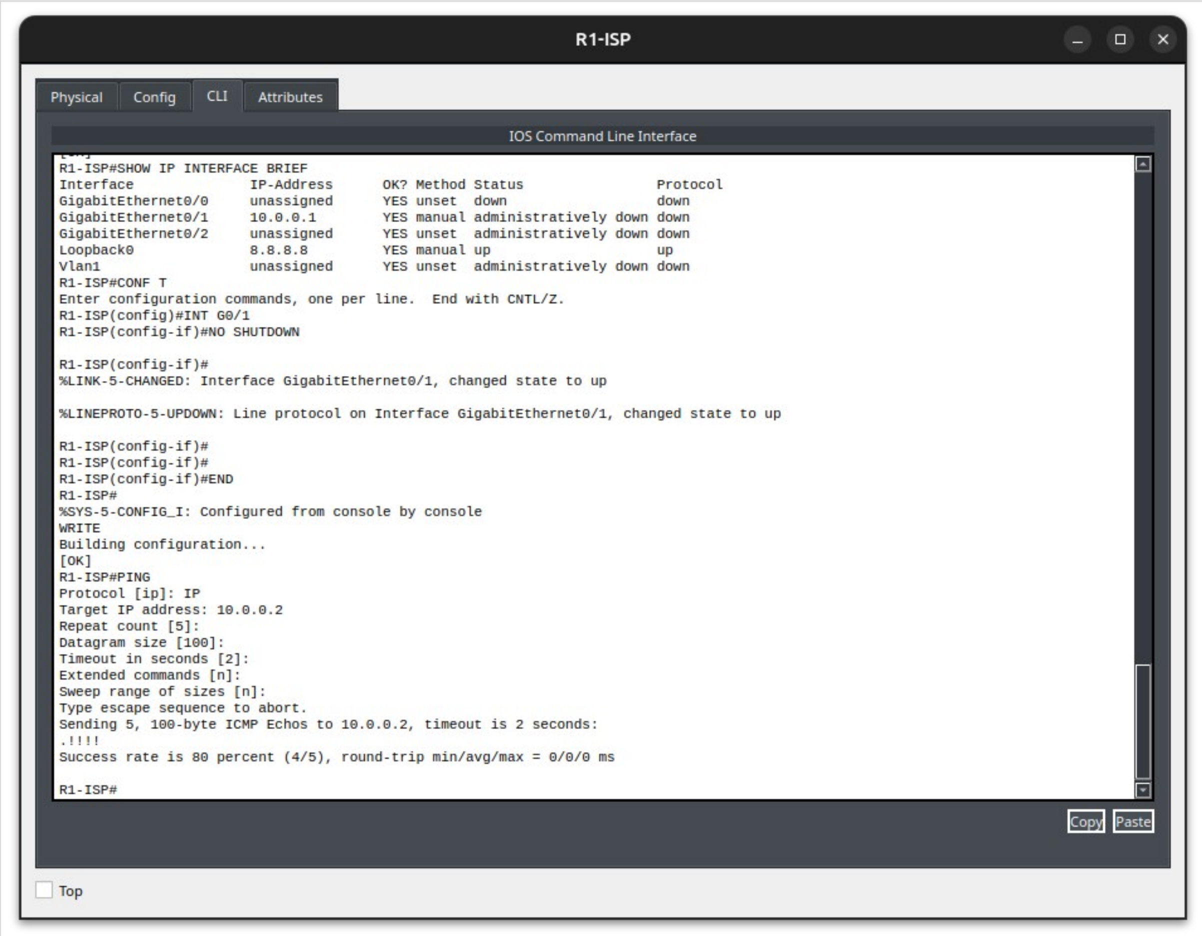


Topologie sous Cisco Packet Tracer.

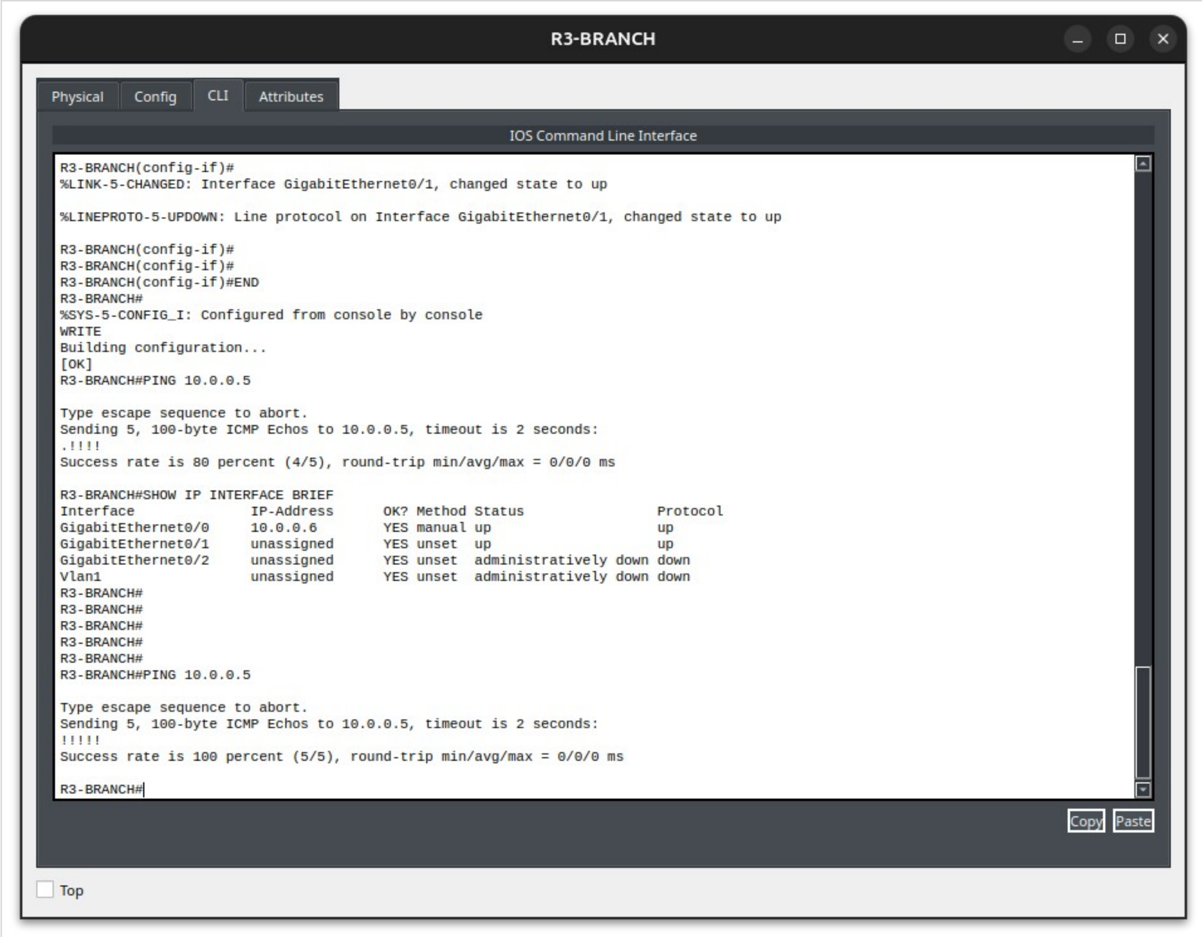
2. Mise en service des liaisons WAN

Activation des interfaces et validation de la connectivité IP de bout en bout sur les liens WAN R1-ISP ↔ R2-HQ (10.0.0.0/30) et R2-HQ ↔ R3-BRANCH (10.0.0.4/30), avant la mise en place d'OSPF.

```
interface g0/1
 no shutdown
ping 10.0.0.2
```



R1-ISP — interface activée, ping vers R2-HQ réussi (80-100 %).

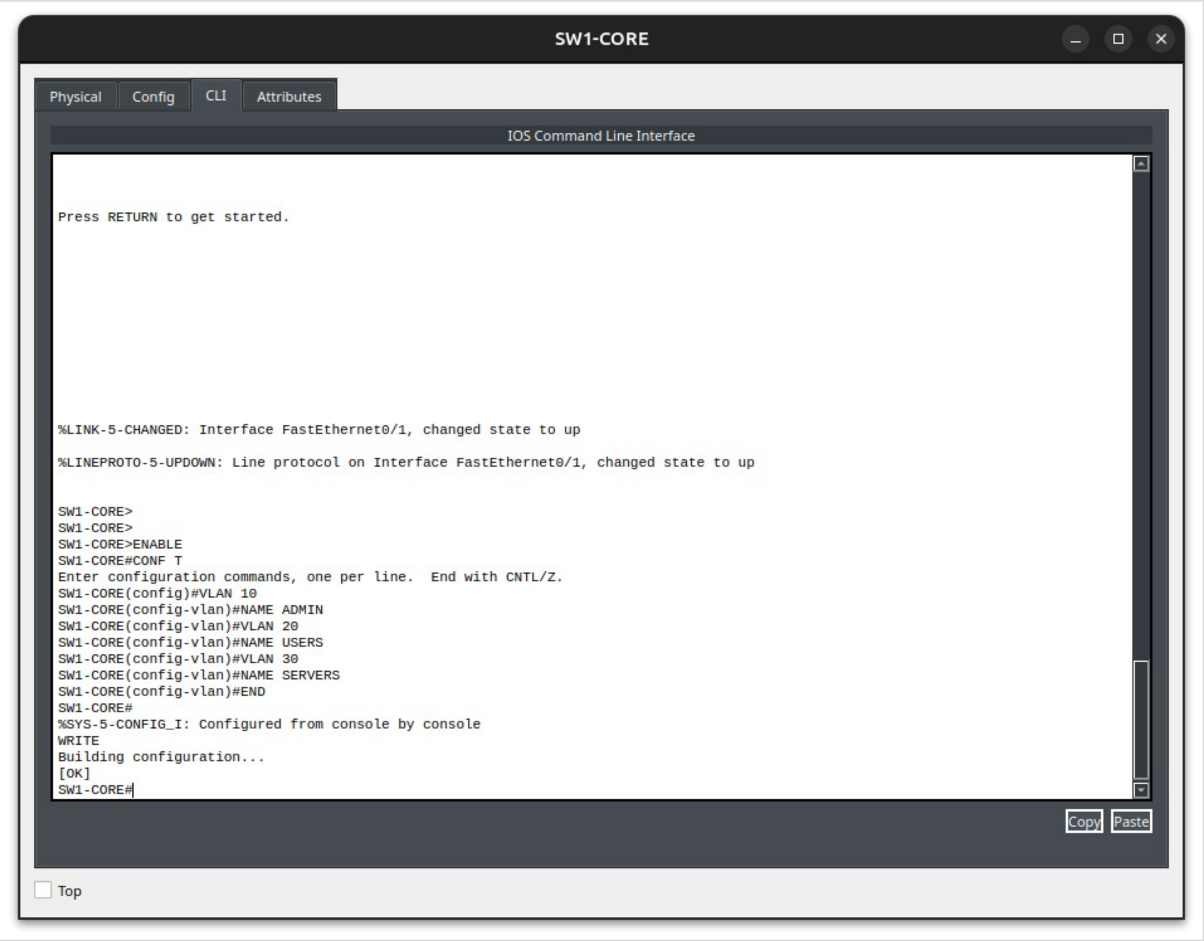


R3-BRANCH — interface up, ping vers R2-HQ réussi (100 %).

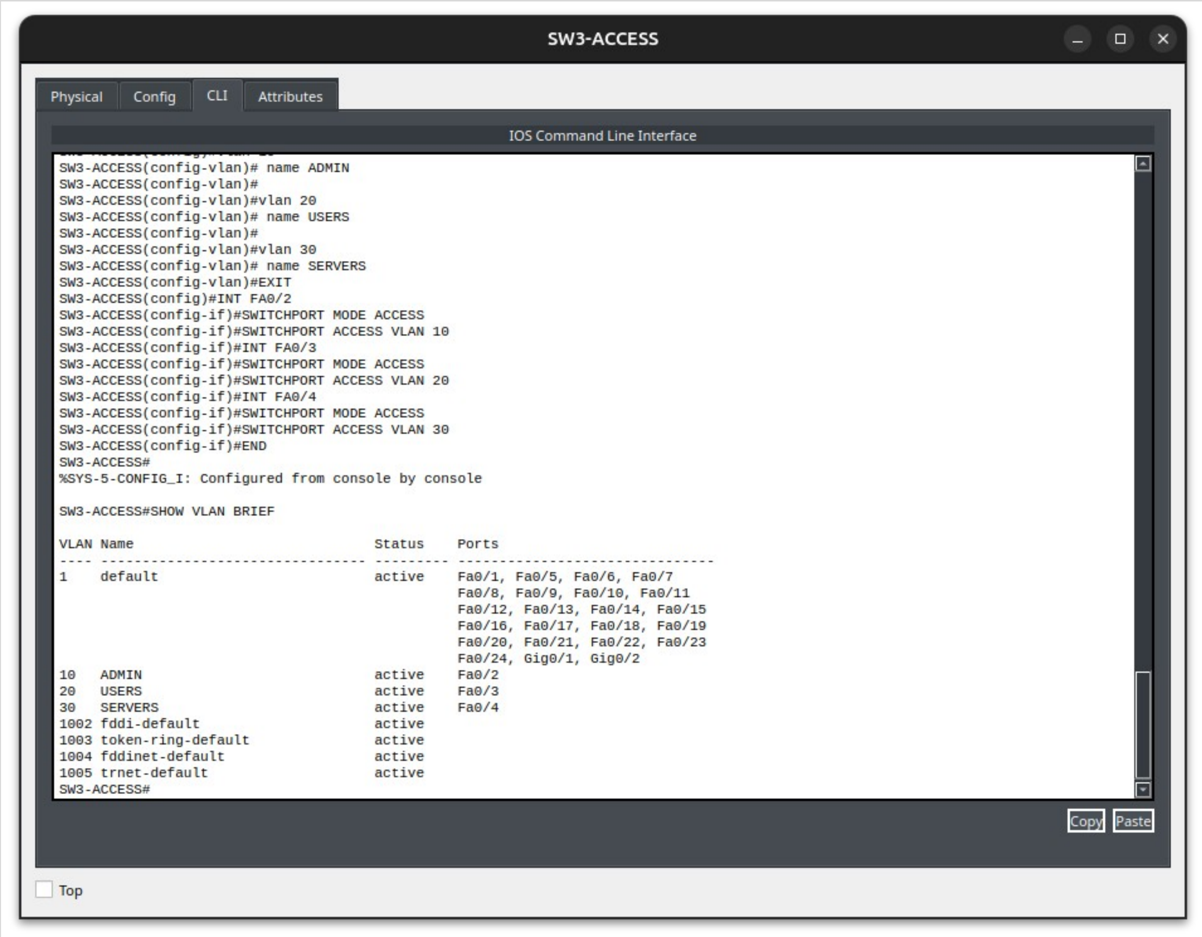
3. VLAN — siège (SW1-CORE / SW3-ACCESS)

Création des VLAN 10 (ADMIN), 20 (USERS) et 30 (SERVERS) sur le switch cœur, puis affectation des ports d'accès correspondants sur SW3-ACCESS.

```
vlan 10
 name ADMIN
vlan 20
 name USERS
vlan 30
 name SERVERS
interface fa0/2
 switchport mode access
 switchport access vlan 10
interface fa0/3
 switchport mode access
 switchport access vlan 20
interface fa0/4
 switchport mode access
 switchport access vlan 30
```



SW1-CORE — création des VLAN 10/20/30.

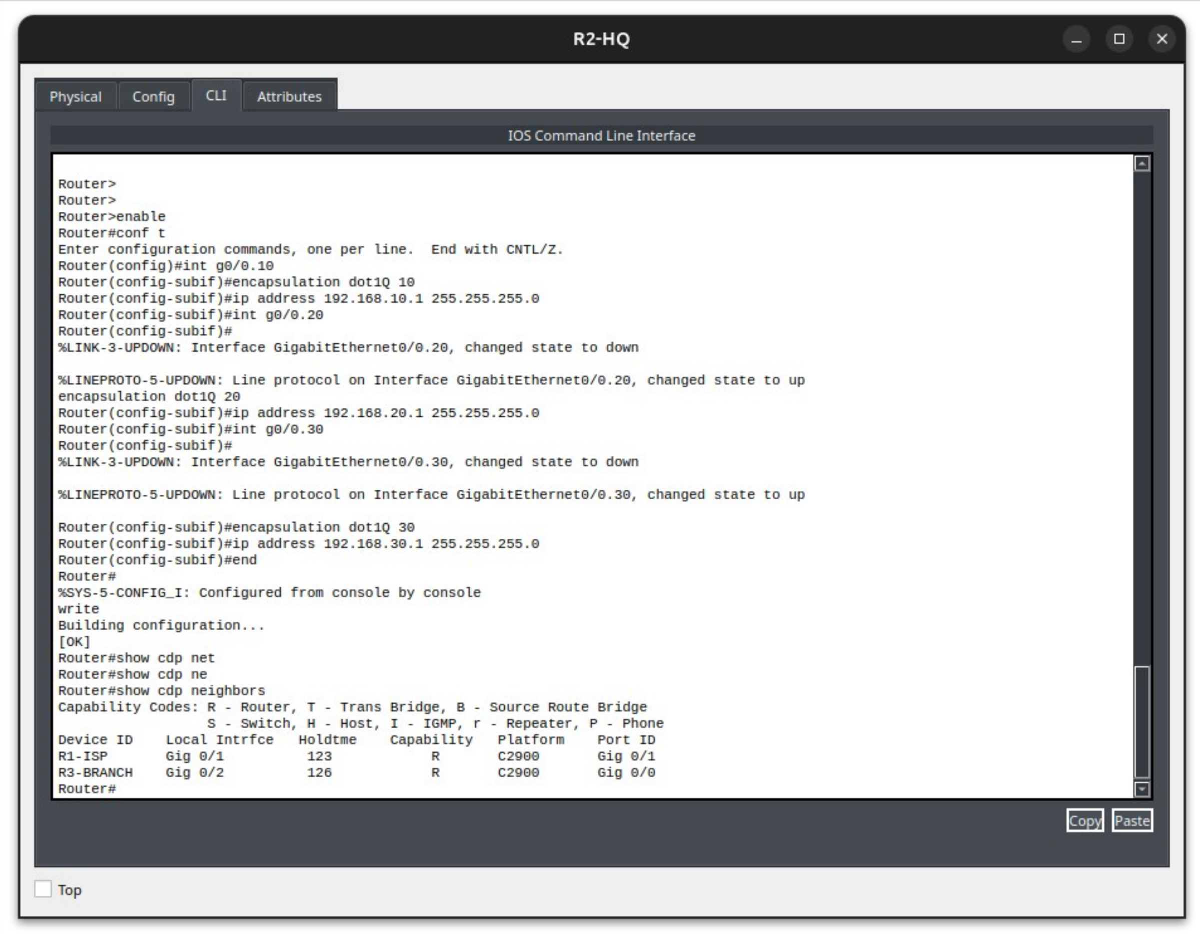


SW3-ACCESS — VLAN nommés et ports d'accès affectés (show vlan brief).

4. Routage inter-VLAN — Router-on-a-Stick (R2-HQ)

Création des sous-interfaces sur R2-HQ (une par VLAN) pour assurer le routage inter-VLAN depuis une interface physique unique en trunk vers SW1-CORE.

```
interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
interface g0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
```

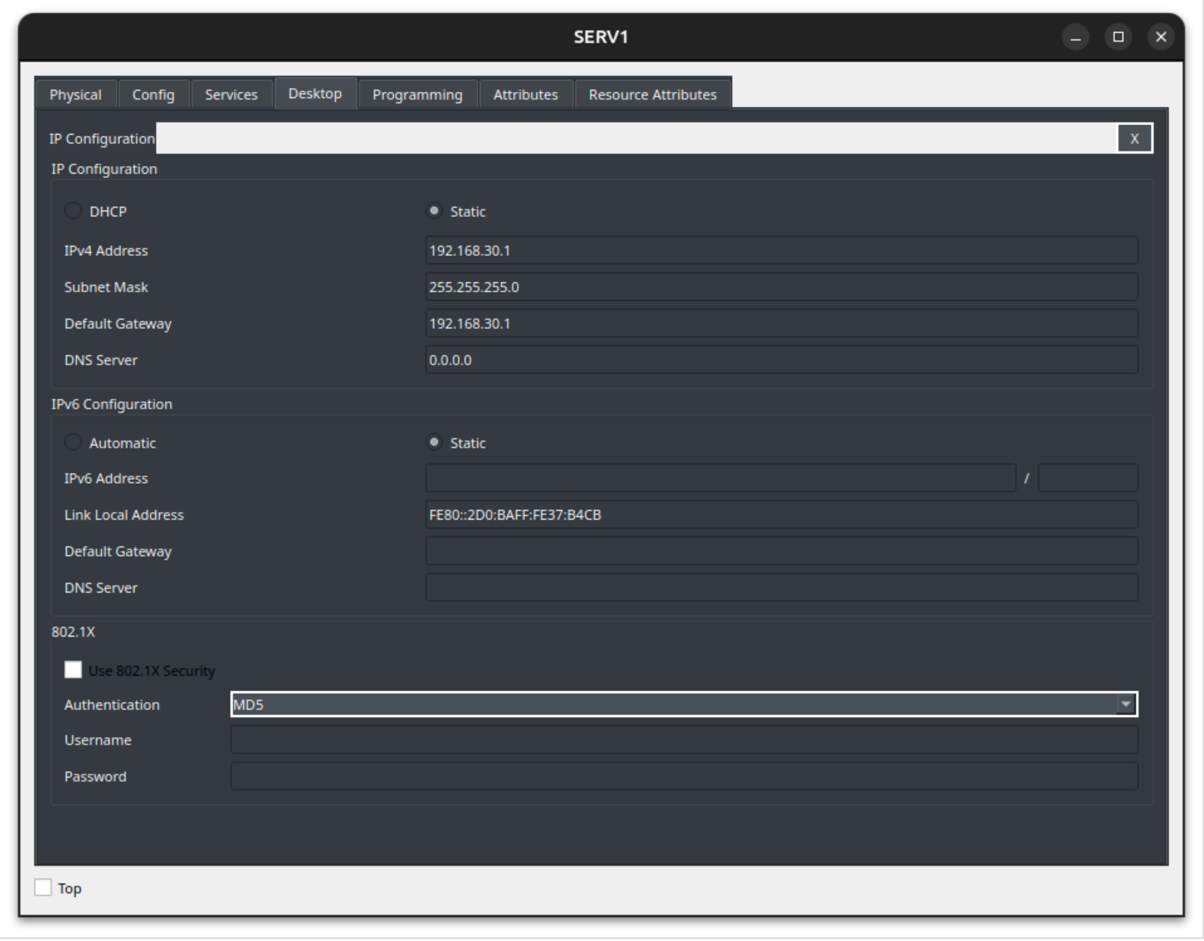


R2-HQ — sous-interfaces VLAN 10/20/30 et voisins CDP (R1-ISP, R3-BRANCH).

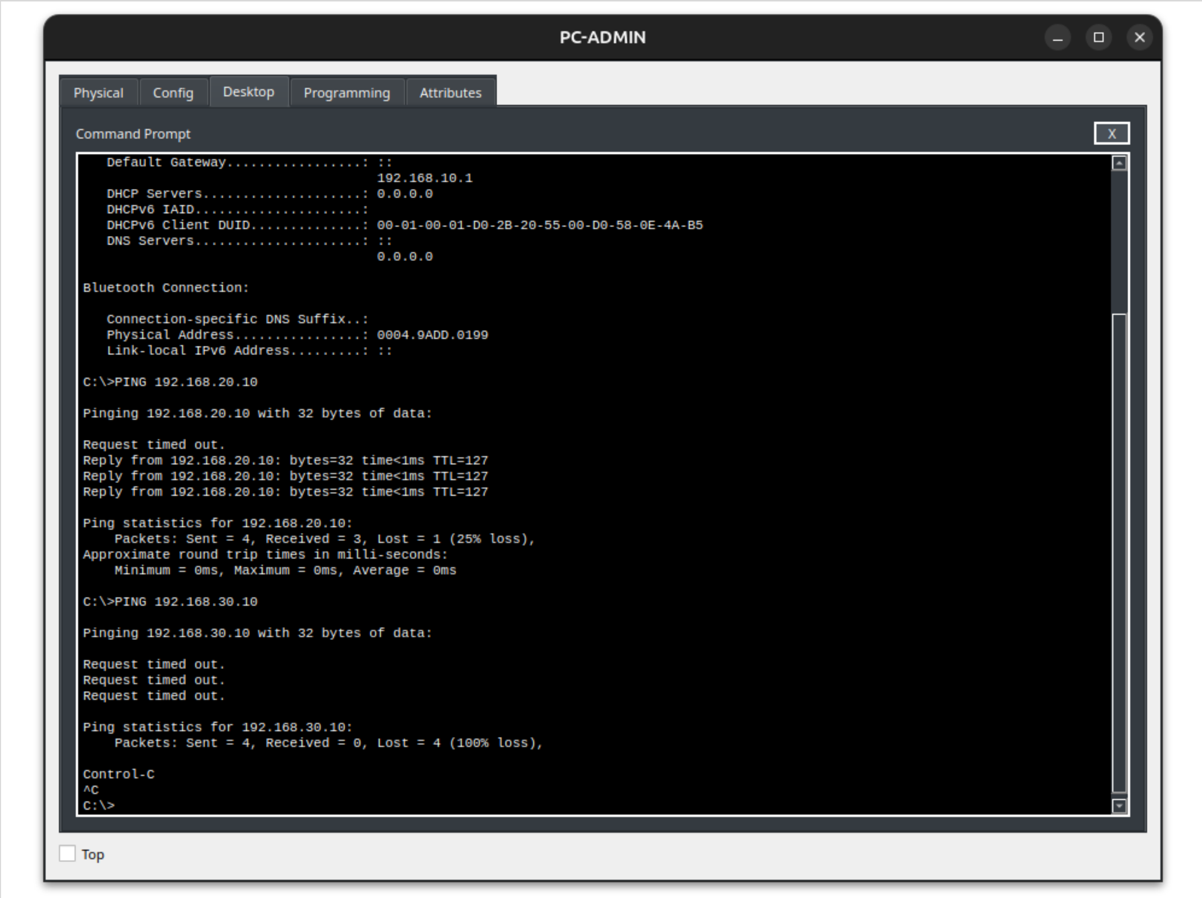
5. Adressage IP statique — serveur SERV1

Attribution d'une adresse IP statique au serveur du VLAN 30 (SERVERS). Une erreur a été commise lors de l’attribution de l’adresse IP : j’ai initialement attribué au serveur l’adresse IP de la passerelle par défaut. De ce fait, le test de connectivité ne fonctionne pas lors du premier essai, comme le montre la capture d’écran ci-dessous.

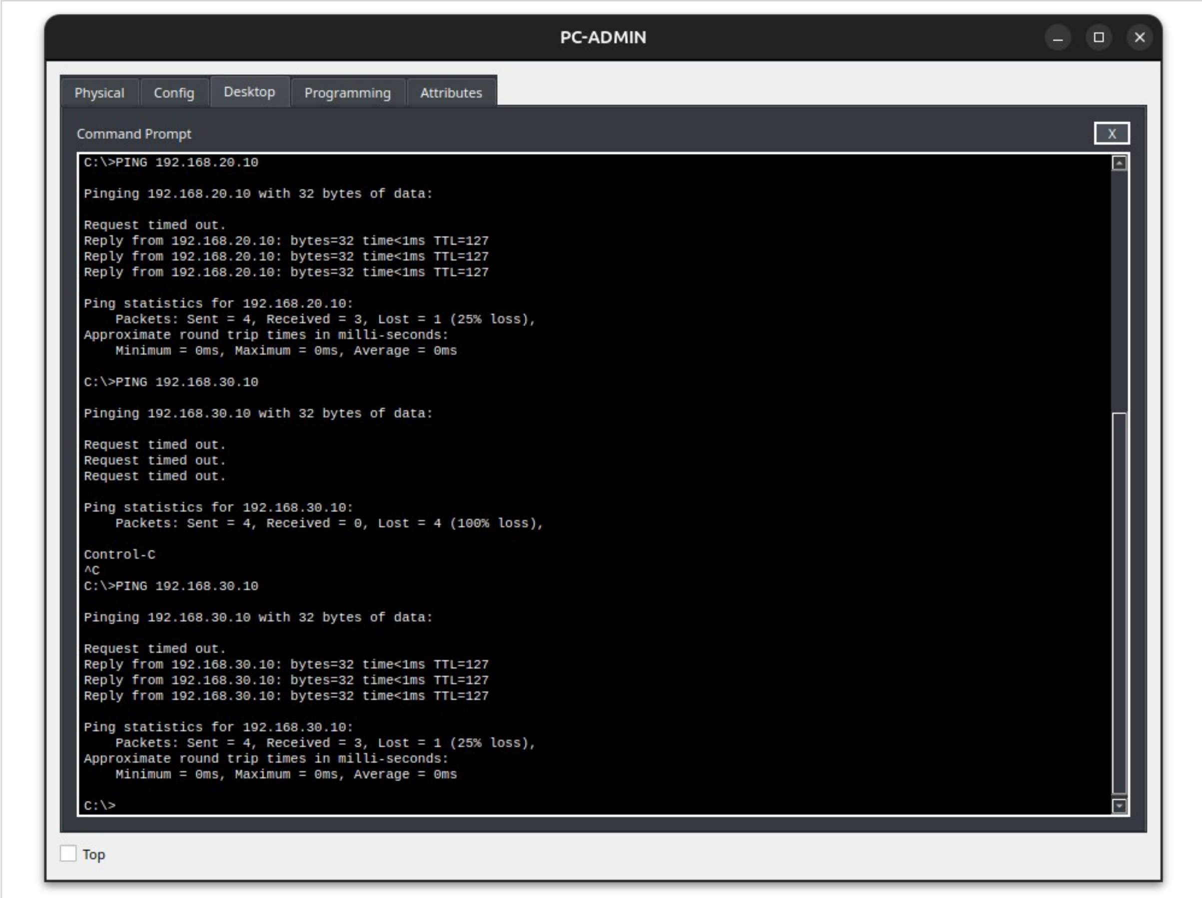
Après avoir constaté cette erreur à la suite du premier test de ping, je la corrige en réattribuant au serveur l’adresse IP **192.168.30.10/24**.



Configuration IP statique de SERV1 — 192.168.30.1/24.



Premier test de ping — Fonctionne vers Laptop-USERS mais pas vers le SERV1 pour cause de mauvaise attribution d’adresse IP.

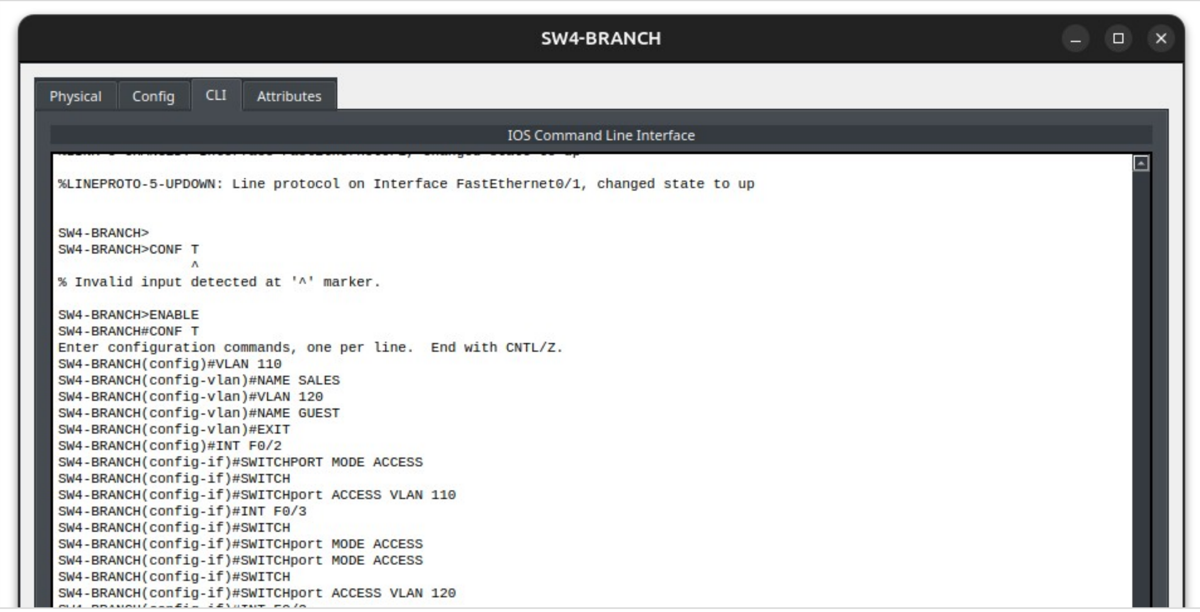


Second test — Après modification de l’adresse, Ping vers SERV1 réussi.

6. VLAN et ports — SW4-BRANCH (agence)

Création des VLAN 110 (SALES) et 120 (GUEST) et affectation des ports d'accès.

```
vlan 110
 name SALES
vlan 120
 name GUEST
interface fa0/2
 switchport mode access
 switchport access vlan 110
interface fa0/3
 switchport mode access
 switchport access vlan 120
```

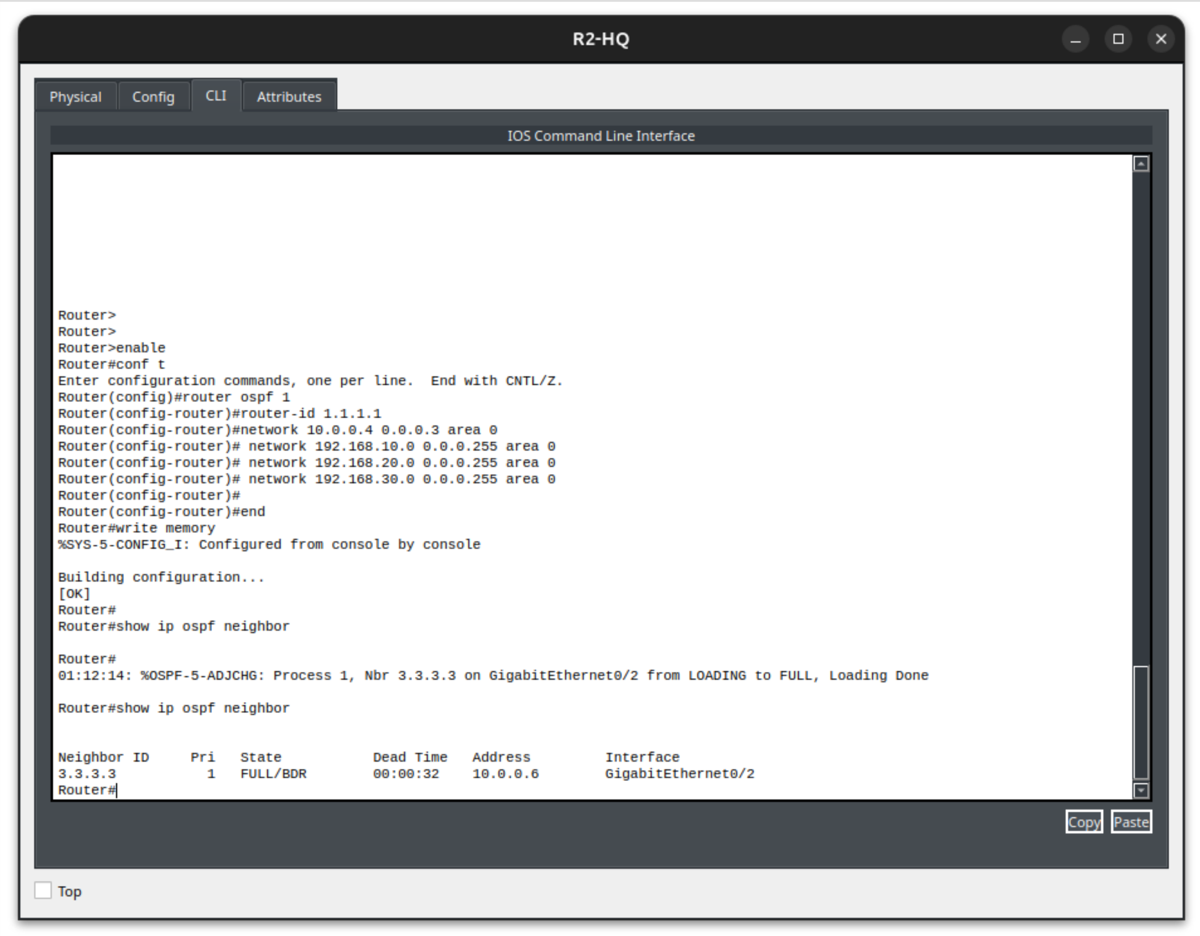


Configuration des VLAN et des ports d'accès sur SW4-BRANCH.

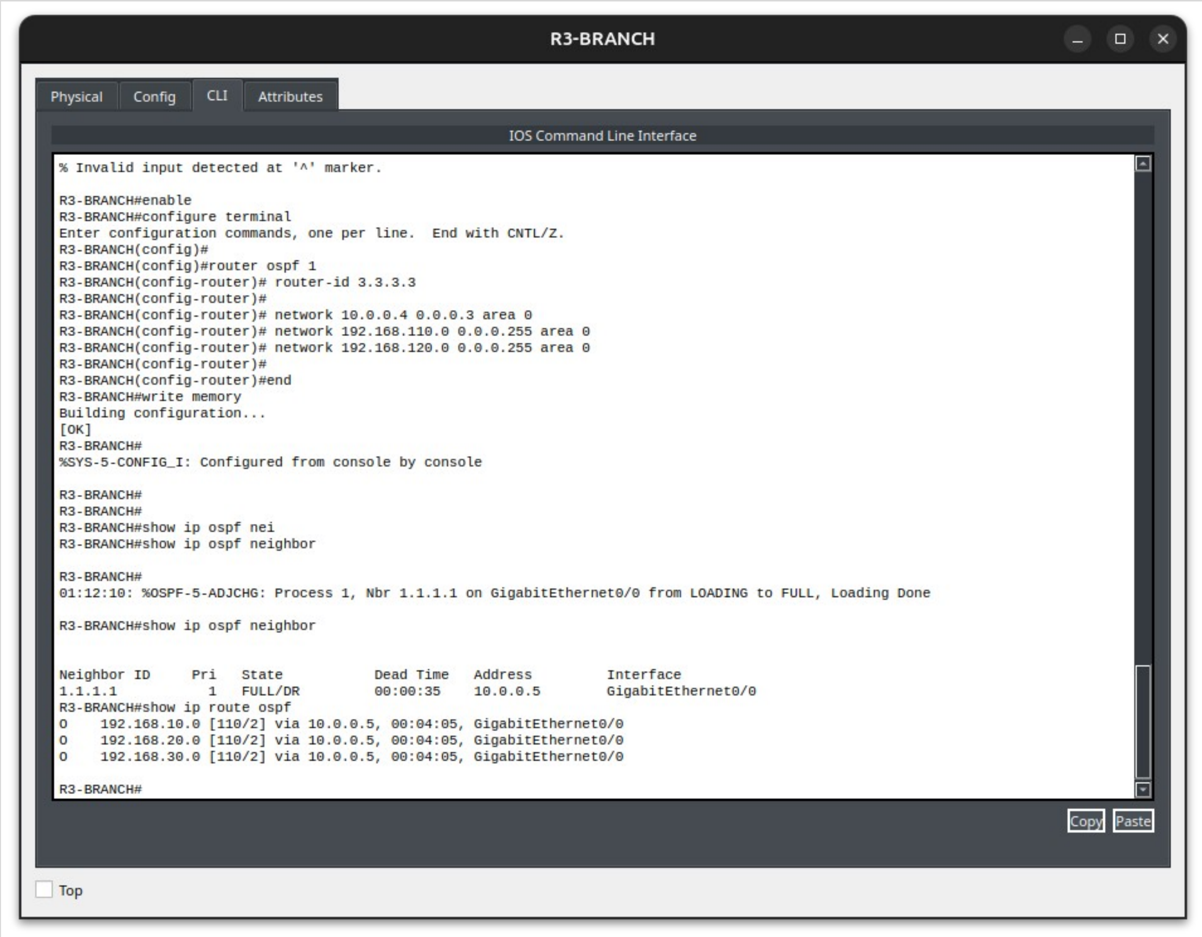
7. Routage dynamique OSPF

Établissement de la relation de voisinage OSPF entre R2-HQ et R3-BRANCH sur le lien WAN 10.0.0.4/30.

```
router ospf 1
 router-id 1.1.1.1
 network 10.0.0.4 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
```



R2-HQ — configuration OSPF, voisinage FULL confirmé.

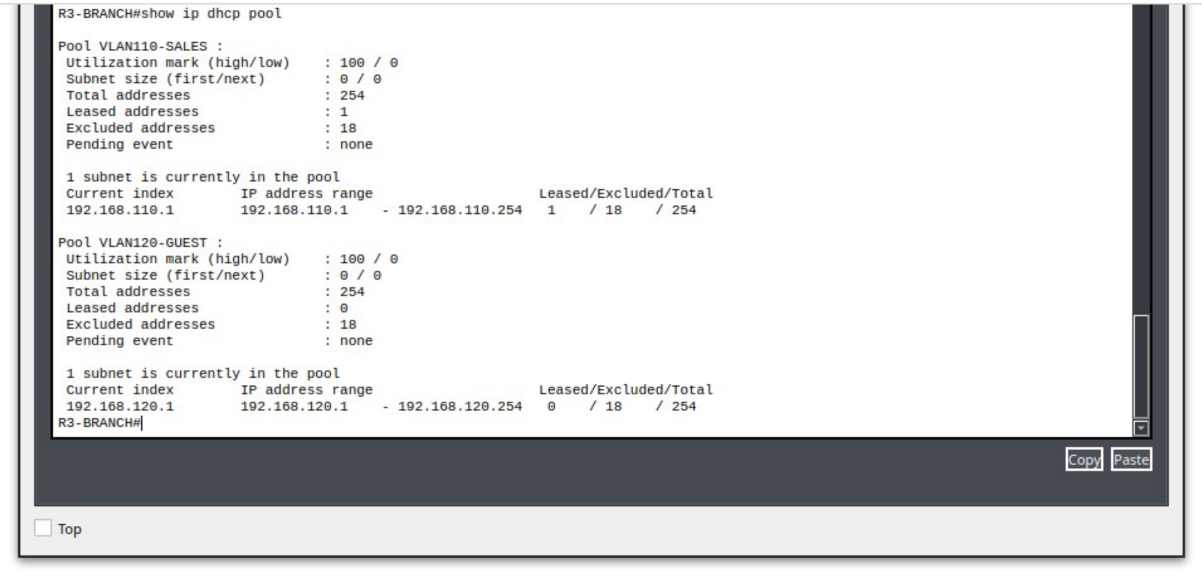


R3-BRANCH — voisinage FULL et routes OSPF apprises (O).

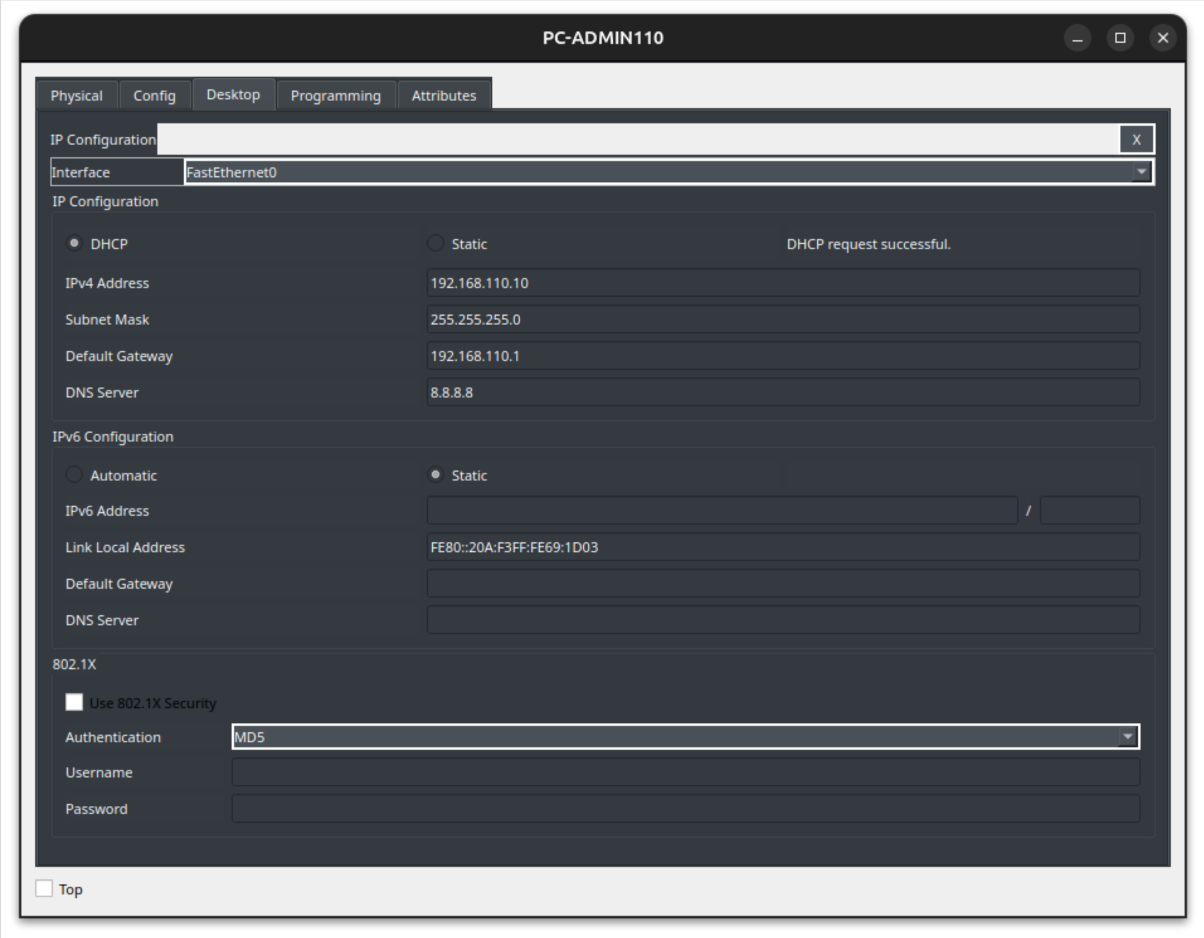
8. Service DHCP — agence

Distribution automatique des adresses pour les VLAN 110 et 120 depuis R3-BRANCH.

```
ip dhcp excluded-address 192.168.110.1 192.168.110.9
ip dhcp pool VLAN110-SALES
 network 192.168.110.0 255.255.255.0
 default-router 192.168.110.1
 dns-server 8.8.8.8
```



Pools DHCP configurés sur R3-BRANCH (SALES et GUEST).

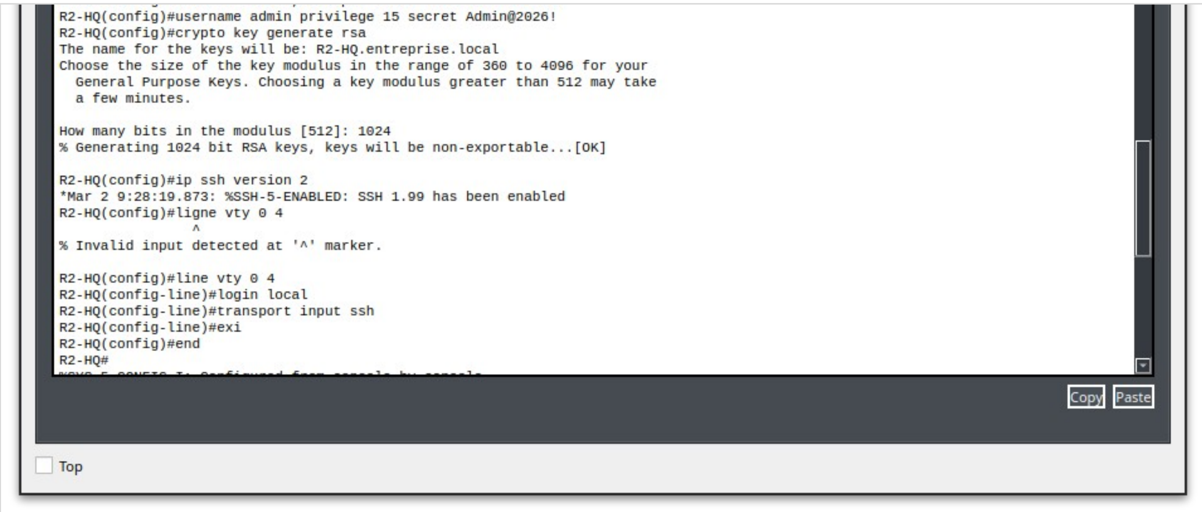


Poste client — bail DHCP obtenu avec succès (VLAN 110).

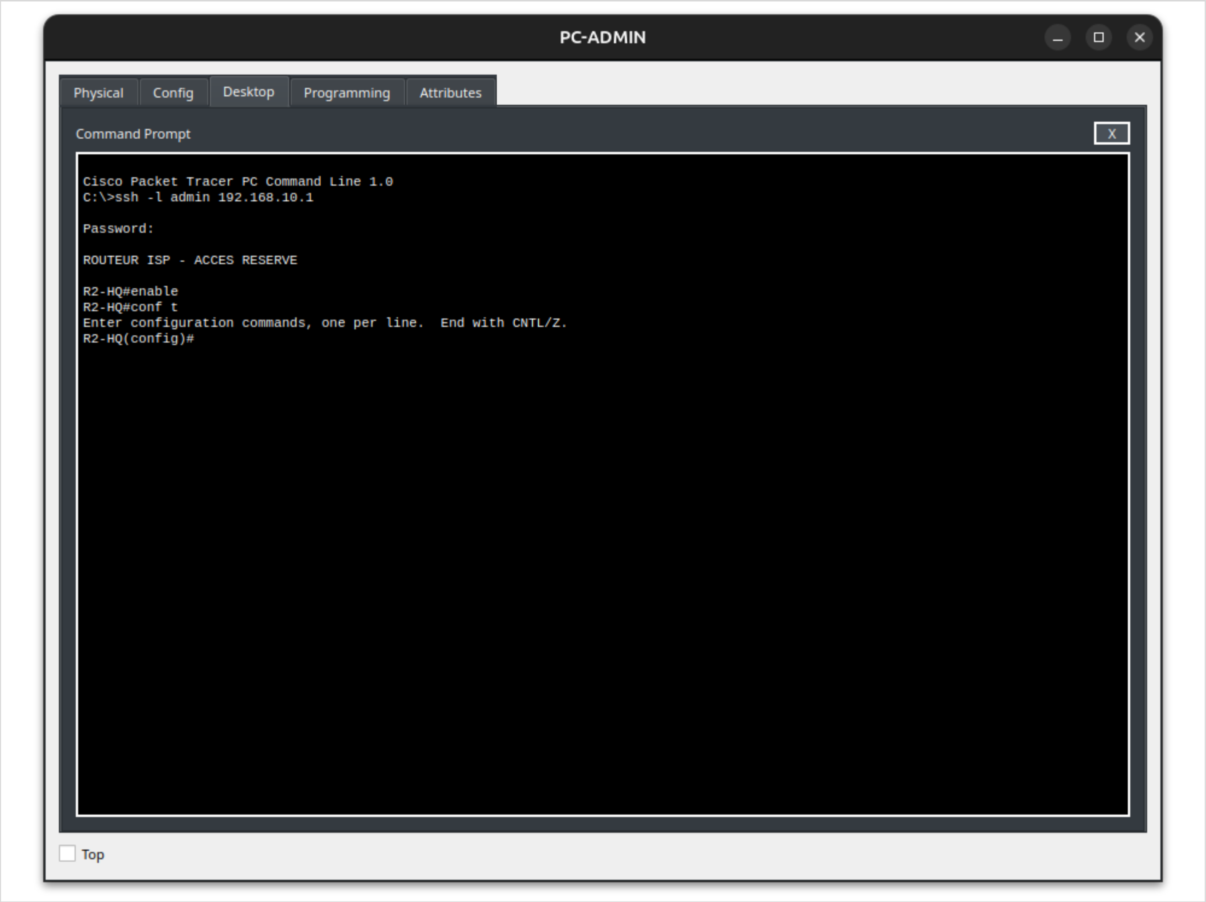
9. Administration à distance SSH (R2-HQ uniquement)

SSH activé uniquement sur R2-HQ pour ce projet ; R3-BRANCH et les switches restent administrés en local/console.

```
hostname R2-HQ
ip domain-name entreprise.local
username admin privilege 15 secret Admin@2026!
crypto key generate rsa
ip ssh version 2
line vty 0 4
 login local
 transport input ssh
```



R2-HQ — activation de SSH v2 et authentification locale.

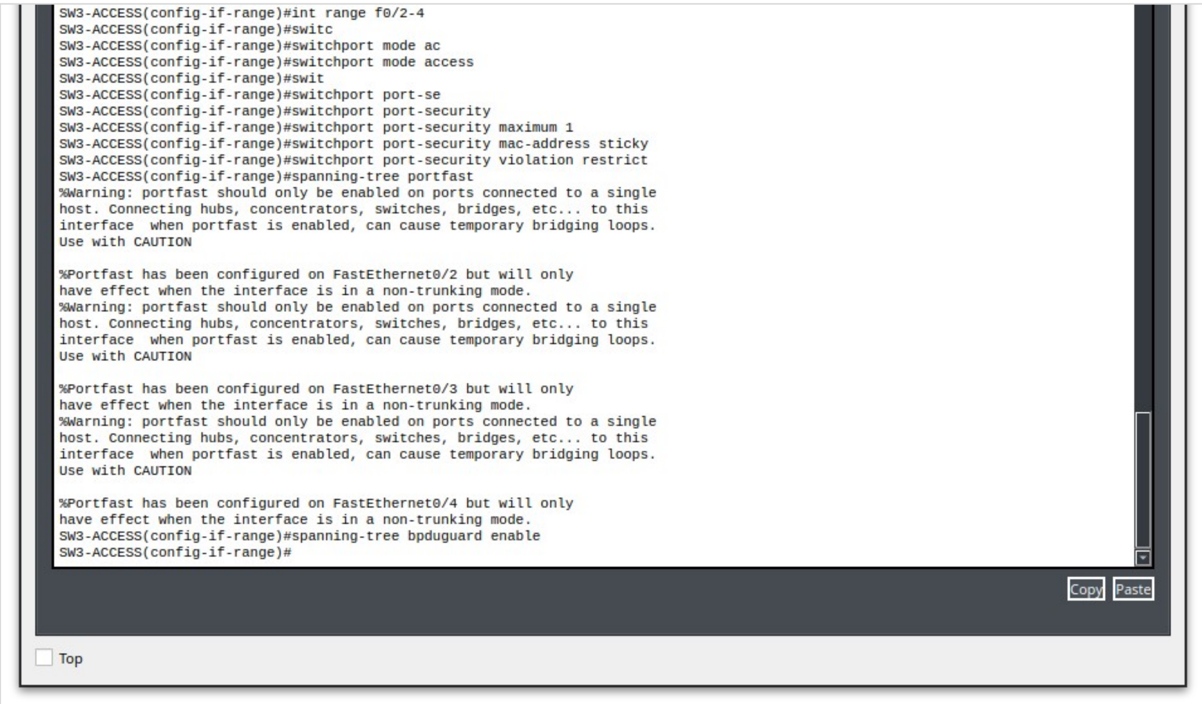


Connexion SSH établie depuis PC-ADMIN vers R2-HQ.

10. Sécurité des ports — Port Security (SW3-ACCESS)

Limitation à une seule adresse MAC par port utilisateur, avec protection BPDU Guard.

```
interface range fa0/2 - 4
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 spanning-tree portfast
 spanning-tree bpduguard enable
```

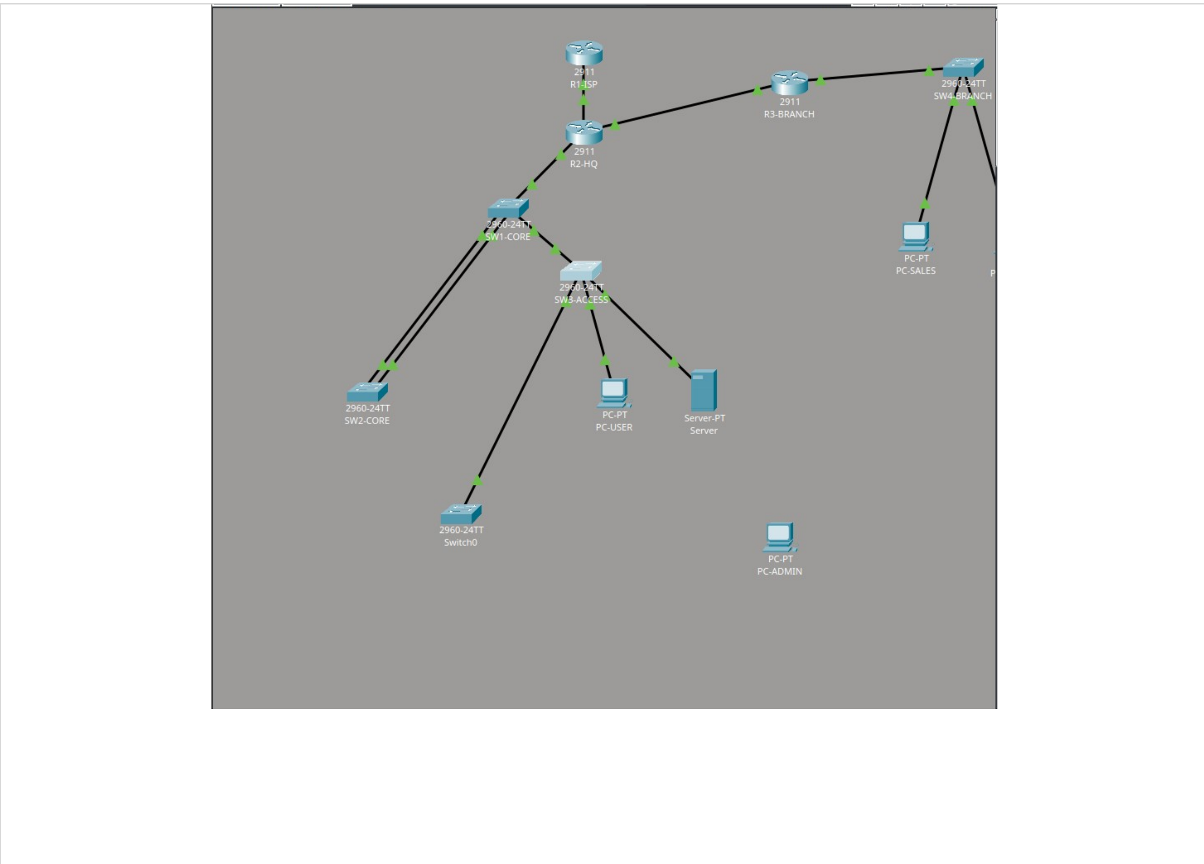


```
SW3-ACCESS(config-if-range)#int range f0/2-4
SW3-ACCESS(config-if-range)#switc
SW3-ACCESS(config-if-range)#switchport mode ac
SW3-ACCESS(config-if-range)#switchport mode access
SW3-ACCESS(config-if-range)#swit
SW3-ACCESS(config-if-range)#switchport port-se
SW3-ACCESS(config-if-range)#switchport port-security
SW3-ACCESS(config-if-range)#switchport port-security maximum 1
SW3-ACCESS(config-if-range)#switchport port-security mac-address sticky
SW3-ACCESS(config-if-range)#switchport port-security violation restrict
SW3-ACCESS(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/4 but will only
have effect when the interface is in a non-trunking mode.
SW3-ACCESS(config-if-range)#spanning-tree bpduguard enable
SW3-ACCESS(config-if-range)#
```

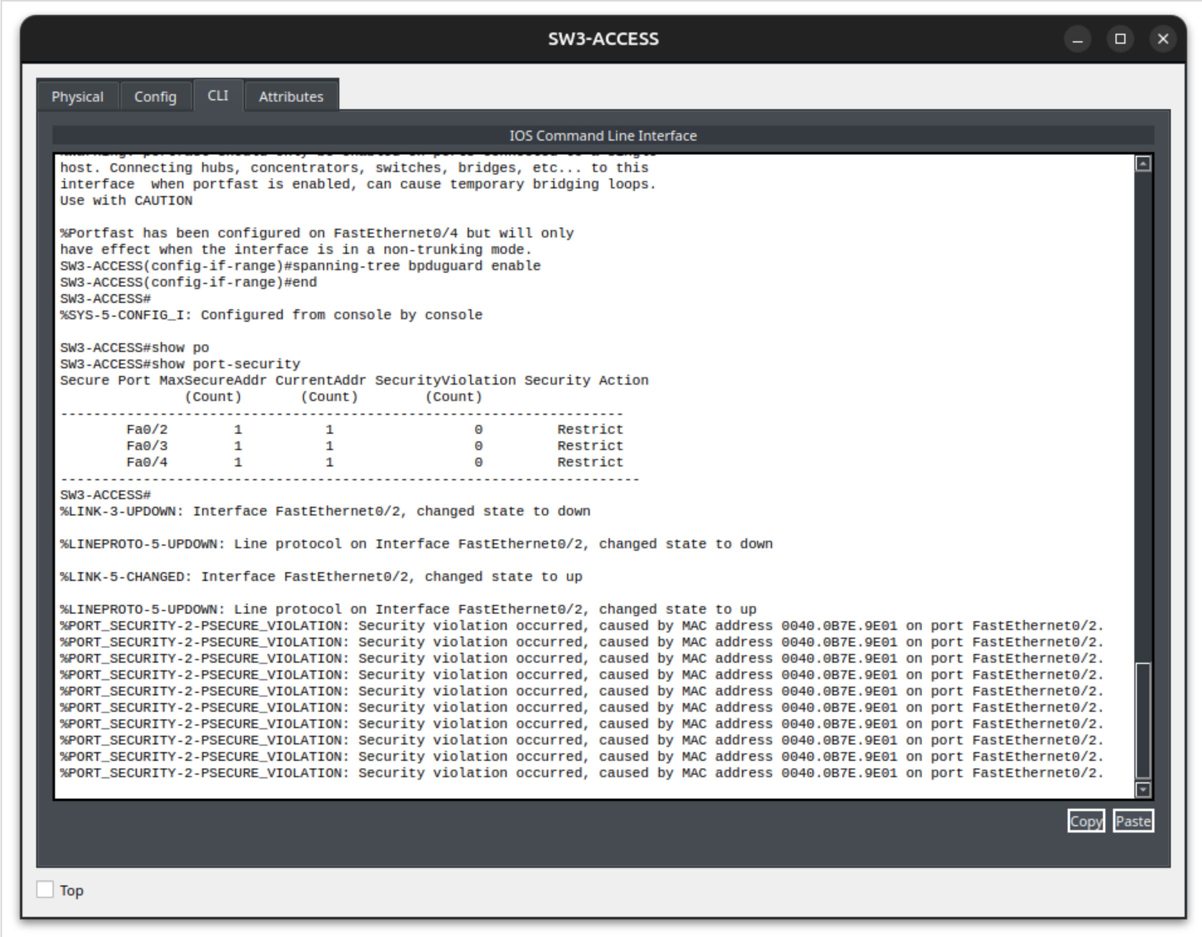

Configuration Port Security appliquée sur SW3-ACCESS.

```
SW3-ACCESS#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
               (Count)       (Count)        (Count)
--------------------------------------------------------------------
        Fa0/2        1           1                0         Restrict
        Fa0/3        1           1                0         Restrict
        Fa0/4        1           1                0         Restrict
----------------------------------------------------------------------
SW3-ACCESS#
```


Vérification — un hôte sécurisé par port, mode restrict.



Test de la protection — un switch non autorisé branché à la place de PC-ADMIN pour déclencher une violation.

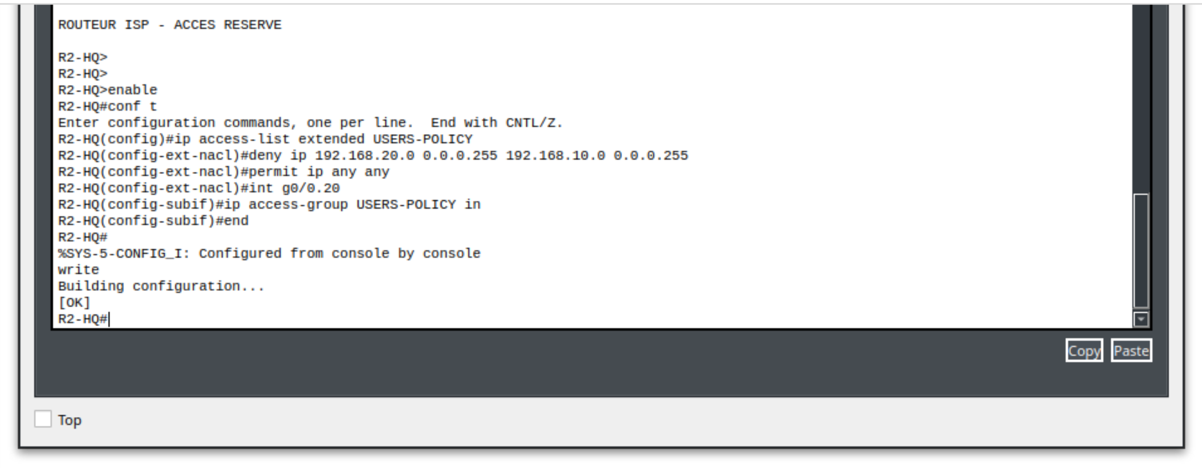


Violation détectée suite au branchement de cet équipement non autorisé.

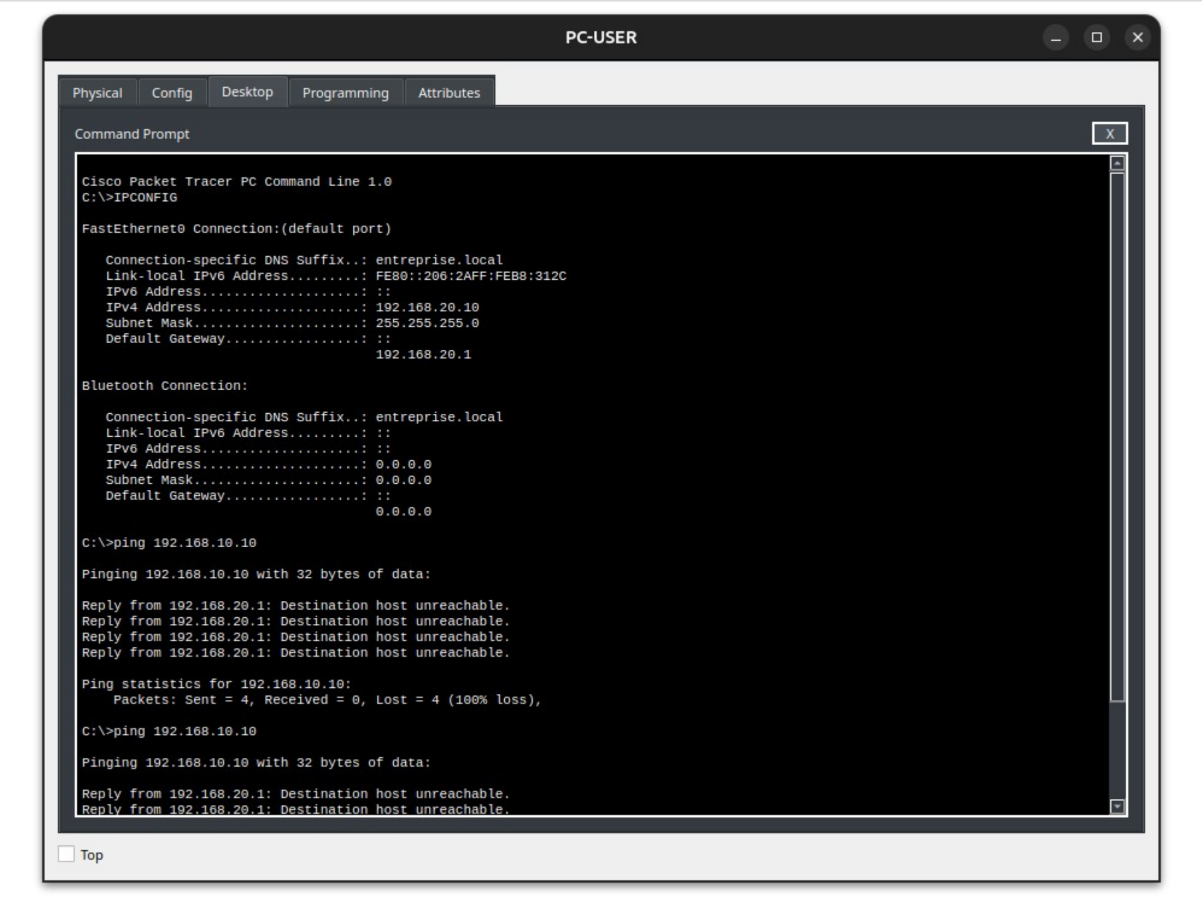
11. ACL inter-VLAN — politique USERS-POLICY

Le VLAN 20 (USERS) est bloqué vers le VLAN 10 (ADMIN) mais garde accès aux serveurs et à l'agence.

```
ip access-list extended USERS-POLICY
 remark BLOCK USERS TO ADMIN
 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 permit ip any any
interface gigabitEthernet 0/0.20
 ip access-group USERS-POLICY in
```



R2-HQ — application de l'ACL sur la sous-interface VLAN 20.



PC-USER — ping vers le VLAN ADMIN bloqué comme attendu.

Annexe — Récapitulatif de toutes les commandes

Toutes les commandes de configuration et de vérification utilisées dans ce projet.

```
interface g0/1
 no shutdown
ping <ip-voisin>
show ip interface brief
```

VLAN (SW1-CORE / SW3-ACCESS)

```
vlan 10
 name ADMIN
vlan 20
 name USERS
vlan 30
 name SERVERS
interface fa0/2
 switchport mode access
 switchport access vlan 10
interface fa0/3
 switchport mode access
 switchport access vlan 20
interface fa0/4
 switchport mode access
 switchport access vlan 30
show vlan brief
```

VLAN (SW4-BRANCH — agence)

```
vlan 110
 name SALES
vlan 120
 name GUEST
interface fa0/2
 switchport mode access
 switchport access vlan 110
interface fa0/3
 switchport mode access
 switchport access vlan 120
```

Trunk 802.1Q

```
interface fa0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
show interfaces trunk
```

EtherChannel LACP

```
interface range fa0/23 - 24
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 channel-group 1 mode active
interface port-channel 1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
show etherchannel summary
```

Routage inter-VLAN — Router-on-a-Stick (R2-HQ)

```
interface g0/0.10
```

```
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
interface g0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
interface g0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
show ip interface brief
```

DHCP (R3-BRANCH — agence)

```
ip dhcp excluded-address 192.168.110.1 192.168.110.9
ip dhcp pool VLAN110-SALES
 network 192.168.110.0 255.255.255.0
 default-router 192.168.110.1
 dns-server 8.8.8.8
 domain-name entreprise.local
ip dhcp excluded-address 192.168.120.1 192.168.120.9
ip dhcp pool VLAN120-GUEST
 network 192.168.120.0 255.255.255.0
 default-router 192.168.120.1
 dns-server 8.8.8.8
 domain-name guest.entreprise.local
show ip dhcp binding
show ip dhcp pool
```

OSPF (R2-HQ / R3-BRANCH)

```
router ospf 1
 router-id 1.1.1.1
 network 10.0.0.4 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
show ip ospf neighbor
show ip route ospf
```

Port Security (SW3-ACCESS)

```
interface range fa0/2 - 4
 switchport mode access
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 spanning-tree portfast
 spanning-tree bpduguard enable
show port-security
show port-security interface fa0/2
show port-security address
```

DHCP Snooping (SW3-ACCESS)

```
ip dhcp snooping
ip dhcp snooping vlan 10,20,30
interface fa0/1
 ip dhcp snooping trust
interface range fa0/2 - 4
 ip dhcp snooping limit rate 10
show ip dhcp snooping
show ip dhcp snooping binding
```

Dynamic ARP Inspection (SW3-ACCESS)

```
ip arp inspection vlan 10,20
interface fa0/1
 ip arp inspection trust
interface range fa0/2 - 4
 ip arp inspection limit rate 15
show ip arp inspection
show ip arp inspection interfaces
```

SSH (R2-HQ uniquement)

```
hostname R2-HQ
ip domain-name entreprise.local
username admin privilege 15 secret Admin@2026!
crypto key generate rsa
ip ssh version 2
line vty 0 4
 login local
 transport input ssh
show ip ssh
```

Management des switches (SVI)

```
interface vlan 10
 ip address 192.168.10.2 255.255.255.0
 no shutdown
ip default-gateway 192.168.10.1
```

ACL inter-VLAN — USERS-POLICY (R2-HQ)

```
ip access-list extended USERS-POLICY
 remark BLOCK USERS TO ADMIN
 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
 remark ALLOW USERS TO SERVERS
 permit ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 remark ALLOW USERS TO BRANCH
 permit ip 192.168.20.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip 192.168.20.0 0.0.0.255 192.168.120.0 0.0.0.255
 permit ip any any
interface gigabitEthernet 0/0.20
 ip access-group USERS-POLICY in
show access-lists
```

ACL — GUEST-POLICY (R3-BRANCH)

```
ip access-list extended GUEST-POLICY
 deny ip 192.168.120.0 0.0.0.255 192.168.10.0 0.0.0.255
 deny ip 192.168.120.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny ip 192.168.120.0 0.0.0.255 192.168.30.0 0.0.0.255
 deny ip 192.168.120.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip any any
interface gigabitEthernet 0/1.120
 ip access-group GUEST-POLICY in
```

NAT/PAT (R2-HQ)

```
interface g0/0.10
 ip nat inside
interface g0/0.20
 ip nat inside
interface g0/0.30
 ip nat inside
interface g0/1
```

```
 ip nat outside
ip access-list standard NAT-HQ
 permit 192.168.10.0 0.0.0.255
 permit 192.168.20.0 0.0.0.255
 permit 192.168.30.0 0.0.0.255
ip nat inside source list NAT-HQ interface g0/1 overload
ip route 0.0.0.0 0.0.0.0 10.0.0.1
show ip nat translations
show ip nat statistics
```

Internet simulé (R1-ISP)

```
interface loopback0
 ip address 8.8.8.8 255.255.255.255
ping 8.8.8.8
```

Sauvegarde de la configuration

```
copy running-config startup-config
write memory
```

Toutes les commandes de vérification (show)

```
show ip interface brief
show vlan brief
show interfaces trunk
show etherchannel summary
show ip route
show ip route ospf
show ip ospf neighbor
show ip ospf
show ip protocols
show ip dhcp binding
show ip dhcp pool
show port-security
show port-security address
show port-security interface fa0/2
show ip dhcp snooping
show ip dhcp snooping binding
show ip arp inspection
show ip arp inspection interfaces
show ip ssh
show access-lists
show ip nat translations
show ip nat statistics
show running-config
```